

Lecture 32/Module 2 & 3

1st and 2nd Order DC Circuits, AC Circuits

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Problem 1

Calculate these complex numbers and express your results in rectangular form:

$$(a) \frac{15 \angle 45^\circ}{3 - j4} + j2$$

$$(b) \frac{8 \angle -20^\circ}{(2 + j)(3 - j4)} + \frac{10}{-5 + j12}$$

$$(c) 10 + (8 \angle 50^\circ)(5 - j12)$$

Answer key

$$\begin{aligned}\text{(a)} \quad \frac{15\angle 45^\circ}{3-j4} + j2 &= \frac{15\angle 45^\circ}{5\angle -53.13^\circ} + j2 \\ &= 3\angle 98.13^\circ + j2 \\ &= -0.4245 + j2.97 + j2 \\ &= \underline{\underline{-0.4243 + j4.97}}\end{aligned}$$

$$\begin{aligned}\text{(b)} \quad \frac{(2+j)(3-j4)}{8\angle -20^\circ} + \frac{10}{-5+j12} &= \frac{6-j8+j3+4}{11.18\angle -26.57^\circ} + \frac{(-5-j12)(10)}{25+144} \\ &= 0.7156\angle 6.57^\circ - 0.2958 - j0.71 \\ &= 0.7109 + j0.08188 - 0.2958 - j0.71 \\ &= \underline{\underline{0.4151 - j0.6281}}\end{aligned}$$

$$\begin{aligned}\text{(c)} \quad 10 + (8\angle 50^\circ)(13\angle -68.38^\circ) &= 10 + 104\angle -17.38^\circ \\ &= \underline{\underline{109.25 - j31.07}}\end{aligned}$$

Problem 2

Find the phasors corresponding to the following signals:

(a) $v(t) = 21 \cos(4t - 15^\circ) \text{ V}$

(b) $i(t) = -8 \sin(10t + 70^\circ) \text{ mA}$

(c) $v(t) = 120 \sin(10t - 50^\circ) \text{ V}$

(d) $i(t) = -60 \cos(30t + 10^\circ) \text{ mA}$

Answer key

$$(a) \quad V = \underline{21 \angle -15^\circ \text{ V}}$$

$$(b) \quad i(t) = 8 \sin(10t + 70^\circ + 180^\circ) = 8 \cos(10t + 70^\circ + 180^\circ - 90^\circ) = 8 \cos(10t + 160^\circ)$$

$$I = \underline{8 \angle 160^\circ \text{ mA}}$$

$$(c) \quad v(t) = 120 \sin(10^3 t - 50^\circ) = 120 \cos(10^3 t - 50^\circ - 90^\circ)$$

$$V = \underline{120 \angle -140^\circ \text{ V}}$$

$$(d) \quad i(t) = -60 \cos(30t + 10^\circ) = 60 \cos(30t + 10^\circ + 180^\circ)$$

$$I = \underline{60 \angle 190^\circ \text{ mA}}$$

Problem 3

Let $\mathbf{X} = 8\angle 40^\circ$ and $\mathbf{Y} = 10\angle -30^\circ$. Evaluate the following quantities and express your results in polar form:

(a) $(\mathbf{X} + \mathbf{Y})\mathbf{X}^*$

(b) $(\mathbf{X} - \mathbf{Y})^*$

(c) $(\mathbf{X} + \mathbf{Y})/\mathbf{X}$

Answer key

Let $\mathbf{X} = 8\angle 40^\circ$ and $\mathbf{Y} = 10\angle -30^\circ$. Evaluate the following quantities and express your results in polar form.

$$(\mathbf{X} + \mathbf{Y})/\mathbf{X}^*$$

$$(\mathbf{X} - \mathbf{Y})^*$$

$$(\mathbf{X} + \mathbf{Y})/\mathbf{X}$$

$$\mathbf{X} = 6.128 + j5.142; \quad \mathbf{Y} = 8.66 - j5$$

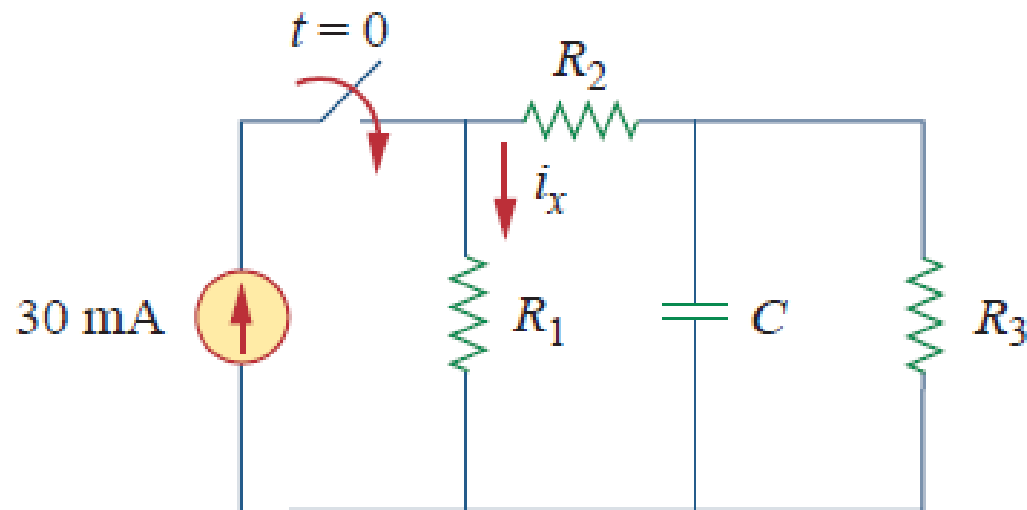
$$\begin{aligned} \text{(a)} \quad (\mathbf{X} + \mathbf{Y})\mathbf{X}^* &= (14.788 + j0.142)(8\angle -40^\circ) \\ &= (14.789\angle 0.55^\circ)(8\angle -40^\circ) = 118.31\angle -39.45^\circ \\ &= \underline{\underline{91.36 - j75.17}} \end{aligned}$$

$$\text{(b)} \quad (\mathbf{X} - \mathbf{Y})^* = (-2.532 + j10.142)^* = \underline{\underline{-2.532 - j10.142}} = 10.453\angle -104.02^\circ$$

$$\begin{aligned} \text{(c)} \quad (\mathbf{X} + \mathbf{Y})/\mathbf{X} &= (14.789\angle 0.55^\circ)/(8\angle 40^\circ) = 1.8486\angle -39.45^\circ \\ &= \underline{\underline{1.4275 - j1.1746}} \end{aligned}$$

Problem 4

*7.50 In the circuit of Fig. 7.117, find i_x for $t > 0$. Let $R_1 = R_2 = 1\text{ k}\Omega$, $R_3 = 2\text{ k}\Omega$, and $C = 0.25\text{ mF}$.



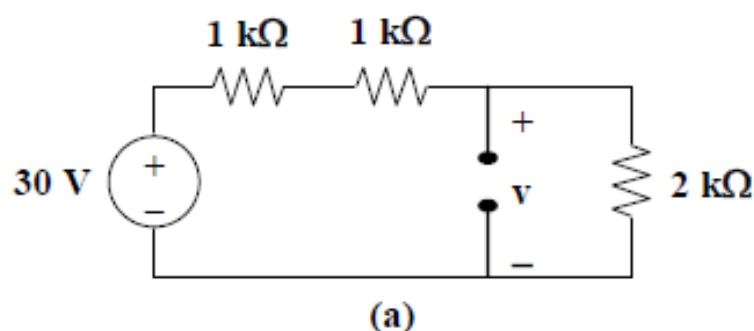
Answer key

For the capacitor voltage,

$$v(t) = v(\infty) + [v(0) - v(\infty)] e^{-t/\tau}$$

$$v(0) = 0$$

For $t > 0$, we transform the current source to a voltage source as shown in Fig. (a).



$$v(\infty) = \frac{2}{2+1+1} (30) = 15 \text{ V}$$

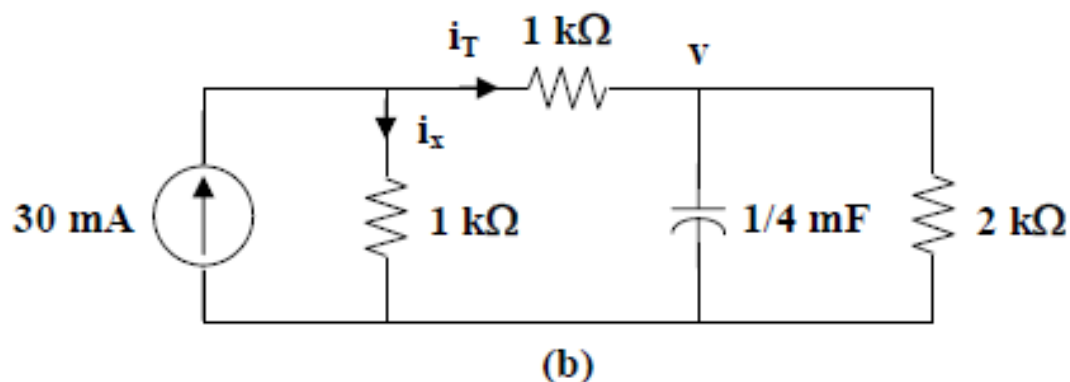
$$R_{th} = (1+1) \parallel 2 = 1 \text{ k}\Omega$$

$$\tau = R_{th} C = 10^3 \times \frac{1}{4} \times 10^{-3} = \frac{1}{4}$$

$$v(t) = 15(1 - e^{-4t}), \quad t > 0$$

Answer key

We now obtain i_x from $v(t)$. Consider Fig. (b).



$$i_x = 30 \text{ mA} - i_T$$

But
$$i_T = \frac{v}{R_3} + C \frac{dv}{dt}$$

$$i_T(t) = 7.5(1 - e^{-4t}) \text{ mA} + \frac{1}{4} \times 10^{-3} (-15)(-4)e^{-4t} \text{ A}$$

$$i_T(t) = 7.5(1 + e^{-4t}) \text{ mA}$$

Thus,

$$i_x(t) = 30 - 7.5 - 7.5e^{-4t} \text{ mA}$$

$$i_x(t) = 7.5(3 - e^{-4t}) \text{ mA}, \quad t > 0$$

Thank You